MIL-STD-1131B

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16 September 1975

# MILITARY STANDARD

## STORAGE SHELF LIFE AND REFORMING PROCEDURES FOR ALUMINUM ELECTROLYTIC FIXED CAPACITORS

FSC 5910

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DEPARTMENT OF DEFENSE
WASHINGTON, D.C. 20360

Storage Shelf Life and Reforming Procedures for Aluminum Electrolytic Fixed Capacitors.

MIL-STD-1131B

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Electronics Technology and Devices Laboratory, ATTN: DELET-DP, Fort Monmouth, NJ 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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1. SCOPE

1.1 Scope. This standard covers the procedures to follow in determining and prolonging the serviceability of aluminum electrolytic fixed capacitors during storage.

1.2 Purpose. The purpose of this document is to establish standards and furnish guidance for the inspection, storage, reformation and issue of aluminum electrolytic fixed capacitors, Federal Supply Class (FSC) 5910.

2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-62 - Capacitors, Fixed, Electrolytic (DC Aluminum, Dry Electrolyte, Polarized) General Specification For.
MIL-C-39018 - Capacitors, Fixed, Electrolytic (Aluminum Oxide), General Specification For.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. DEFINITIONS

3.1 Visual inspection. A visual inspection is an inspection by visual means to observe the item and its packaging and packing to detect deficiencies. Visual inspection normally does not require disassembly or testing of the item.

3.2 Technical inspection. A technical inspection is an inspection by visual or other means such as disassembly, measurement, or performance testing.

4. GENERAL REQUIREMENTS

4.1 Type of storage. Aluminum electrolytic fixed capacitors shall be stored in warehouse space under one of the following conditions:

a. Condition A-
(1) MIL-C-62 and non-military grade - Temperature +35°C or less, 60 ±15 percent relative humidity, and air pressure of 725 ±75 millimeters of mercury.
(2) MIL-C-39018 - Temperature +40°C or less, 50 ±20 percent relative humidity, and air pressure of 725 ±75 millimeters of mercury.
b. Condition B - Temperature, relative humidity, or air pressure other than that specified above, within specified storage limits.

4.2 Storage and issue acceptability. MIL-C-62 and non-military grade aluminum electrolytic fixed capacitors will be acceptable for storage and issue within a 6-year certified period computed from date of manufacture if stored as specified in 4.1 Condition A(1), or within a 4-year period computed from date of manufacture if stored as specified in 4.1 Condition B, or within 3 years from date of last inspection (see 4.3).



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MIL-C-39018 aluminum electrolytic fixed capacitors will be acceptable for storage and issue within a 10-year certified period computed from date of manufacture if stored as specified in 4.1 Condition A(2), or within a 5-year period computed from date of manufacture if stored as specified in 4.1 Condition B, or within 4 years from date of last inspection (see 4.3).

4.3 Inspection frequency. MIL-C-62 and non-military grade - At the end of 6 years or 4 years from the date of manufacture, depending upon storage conditions (see 4.2) or 3 years from the last inspection period, capacitors shall be inspected (see 4.6) to determine if the dc leakage requirements of the applicable procurement document can be met.

MIL-C-39018 - At the end of 10 years or 5 years from the date of manufacture, depending upon storage conditions (see 4.2) or 5 years from the last inspection period, capacitors shall be inspected (see 4.6) to determine if the dc leakage requirements of the applicable procurement document can be met.

4.4 Inspection sampling. At the end of the specified period (see 4.3), capacitors shall be selected in accordance with MIL-STD-105, inspection level II, with an acceptable quality level (AQL) of 2.5 percent and tested as specified (see 4.6). If the number of defectives exceeds those allowed, all capacitors shall be tested; those units not meeting the requirements of the applicable procurement document will be reformed and redated, if acceptable. All units exhibiting out-of-tolerance dc leakage characteristics after reforming shall be discarded.

4.5 Disposal period. MIL-C-62 and non-military grade aluminum electrolytic fixed capacitors more than 12 years old shall be disposed of. MIL-C-39018 aluminum electrolytic fixed capacitors more than 15 years old shall be disposed of.

4.6 Inspection method. The method of inspection shall be visual and technical (see 4.3 and 5.1).

4.7 Defects list. Capacitors shall be inspected for the following defects:

a. Damaged material.
b. Mechanical.
c. Excessive corrosion.
d. Electrical failure - DC leakage.
e. Electrical failure - Out of tolerance.
f. Marking (specify).
g. Preservation, packaging and packing (specify).

5. DETAILED REQUIREMENTS

5.1 DC leakage limits. The dc leakage limits shall be in accordance with the applicable procurement document. If non-military limits are not available, use applicable MIL-C-62 limits.

5.2 Reforming. Groups of electrolytic capacitors shall be reformed in accordance with figure 1.

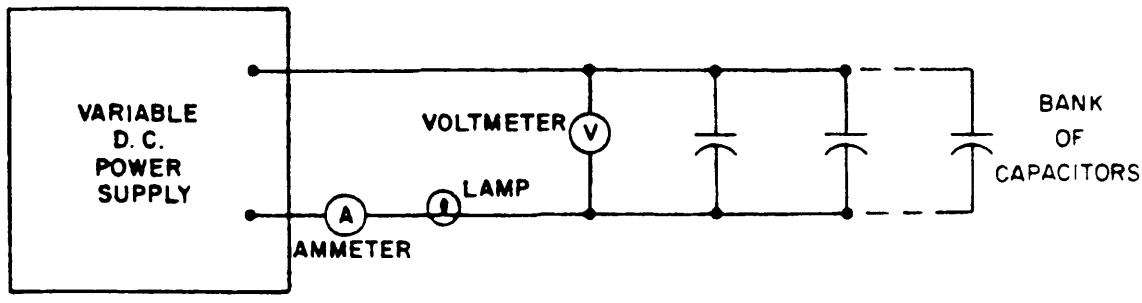



FIGURE 1. Schematic diagram.



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5.2.1 Capacitor reforming groups. The tungsten filament lamp in series with the capacitor bank will serve as a limiting resistor, which allows the voltage across the capacitors to rise as the leakage current decreases. The lamp size shall be chosen so that the load current will not exceed an average of 5 milliamperes (mA) per capacitor. Otherwise, internal heating may cause permanent damage to the capacitors under test. The reforming current for each capacitor or section should be limited to a maximum of 5 mA so the total current for each bank is equal to the number of capacitors or sections being reformed, multiplied by 0.005 amperes.

Example: Determine what size lamp shall be placed in series with a bank of 50 capacitors with similar voltage ratings.

| No. of capacitors | | Amperes | | Amperes |
|---|---|---|---|---|
| 50 | x | .005 | = | 0.250 |

Since the capacitors are initially uncharged, their resistance may be considered negligible and the entire voltage drop would then appear across the lamp. The wattage of the lamp would then be:

| Amperes | | Volts | | Watts |
|---|---|---|---|---|
| 0.250 | x | (rated) | = | ____ |

5.2.2 Capacitor voltage. The power supply output voltage shall be so adjusted that the initial total current does not exceed 5 mA per axial leaded capacitor and 10 mA per radial, threaded terminal capacitor. As the leakage current decreases, the power supply output voltage will be adjusted so that the total current does not exceed current specified above. This procedure will be repeated until rated voltage appears across the capacitor bank or until 4 hours have elapsed.

5.2.3 Capacitor reforming time. The leakage current which an electrolytic capacitor will pass increases with temperature rise. Heat is generated in the electrolytic capacitor during the reforming process. Accordingly, leakage current measurements made immediately after reforming will be higher than at ambient temperature. If leakage current measurements taken immediately after reforming are within specified limits, the capacitor is acceptable. The capacitors should be reformed for a maximum period of 1 hour or for a shorter period where the leakage current measurements fall within specified limits. Where the limits are still exceeded after the 1-hour period, the capacitors should be removed from the voltage source, held for approximately 24 hours, and the leakage test performed again.

5.3 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking shall be in accordance with the applicable procurement document.



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6. NOTES

6.1 Surge voltage. The components of an electronic dc voltage supply will provide a measure of protection to the extent of absorbing a portion of the line surge. Electrolytic capacitors are expected to withstand surge voltage of the magnitudes indicated below.

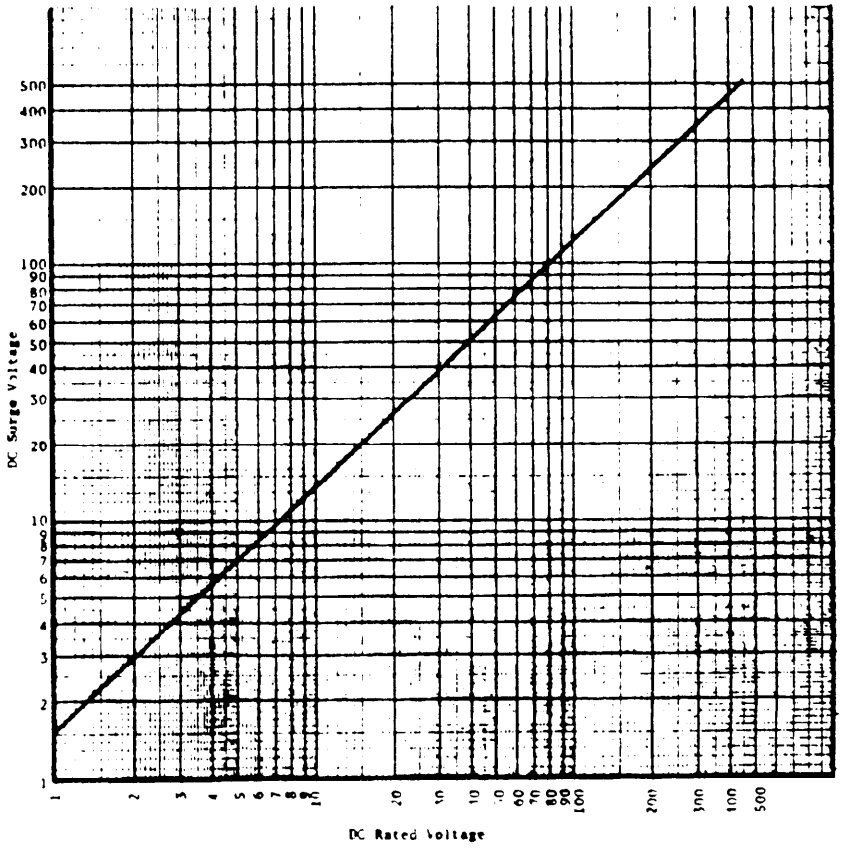



FIGURE 2. Surge voltage.

NOTE: Surge voltages shown are conservative and are given for general field service guidance. Detail specification sheets may permit higher values in some instances. Refer to individual specification sheets for details where necessary.



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| Capacitors rated voltage at equipment ambient temperature | DC surge voltage | Percent surge allowed |
|---|---|---|
| 5 | 7 | 40 |
| 6 | 8 | 33 |
| 7 | 10 | 43 |
| 7.5 | 10 | 33 |
| 10 | 15 | 50 |
| 15 | 20 | 33 |
| 16 | 20 | 25 |
| 20 | 25 | 25 |
| 25 | 30 | 20 |
| 30 | 40 | 33 |
| 40 | 50 | 25 |
| 50 | 60 | 20 |
| 60 | 75 | 25 |
| 75 | 90 | 20 |
| 100 | 125 | 25 |
| 150 | 175 | 16.6 |
| 200 | 225 | 12.5 |
| 225 | 250 | 11.1 |
| 250 | 275 | 10 |
| 275 | 300 | 9.1 |
| 300 | 325 | 8.3 |
| 350 | 375 | 7.1 |
| 400 | 450 | 12.5 |
| 450 | 500 | 11.1 |

6.2 Line voltage variations. It is recommended that a constant voltage transformer or regulator be inserted in the equipment power line where line surges are found to be greater than those outlined above.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army -
Navy -
Air Force - 17
DLA - ES

User activities:
Army -
Navy -
Air Force -

Preparing activity:
Army - ER

Agent:
DLA - ES

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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